• By defn, a pivot position in the matrix corresponds to the leading 1 of a row in reduced row echelon form.

$$\begin{bmatrix} 10 & -3 & -2 \end{bmatrix} \xrightarrow[\times 1/10]{Row1 = Row1} \begin{bmatrix} 1 & -3/10 & -2/10 \end{bmatrix}$$

<u>Verification of whether the final matrix is in rref form.</u>

1. Row 1 is a non-zero row. ∴ The condition that all non-zero rows must be above the zero rows is vacuously True.
2. Row 1 Leading Entry = 1 in Column 1
∴ The condition that each row must have its leading entry in a column to the right of the columns of the leading entries of all rows above it is vacuously True.
3. The condition that for each column having a leading entry, all the positions below the leading entry position must contain 0 is vacuously True.
4. The condition that each leading entry must be 1 is True.
5. The condition that for each column having a leading entry, all the positions above the leading entry position must contain 0 is vacuously True.

∴ The matrix $\begin{bmatrix} 1 & -3/10 & -2/10 \end{bmatrix}$ is in reduced row echelon form.

Columns 2 and 3 have no pivots. ∴ The variables x_2 and x_3 are free.

The corresponding equation is: $x_1 - \frac{3}{10}x_2 - \frac{2}{10}x_3 = 0 \Rightarrow \boxed{x_1 = 0.3x_2 + 0.2x_3}$

∴ The solution set is: (1) $x_1 = x_2 = x_3 = 0$ (2) $x_1 = 0.3x_2 + 0.2x_3$, x_2, x_3 are respectively free. □

As a vector, the general solution is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} .3x_2 + .2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} .3x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} .2x_3 \\ 0 \\ x_3 \end{bmatrix} = x_2 \underbrace{\begin{bmatrix} .3 \\ 1 \\ 0 \end{bmatrix}}_{u} + x_3 \underbrace{\begin{bmatrix} .2 \\ 0 \\ 1 \end{bmatrix}}_{v}$$

(with x_2, x_3 free)

(2)

This calculation shows that every solution of (1) is a linear combination of the vectors u and v, shown in (2). That is, the solution